

GUAN'AO YAN

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RESEARCH AREA

My research interests lie in developing new statistical methods for understanding real-world data. Specific research topics include

- Statistical Bioinformatics
 - Statistical methods for analyzing high-dimensional single-cell and spatial omics data
 - Using synthetic data to enhance the statistical rigor in single-cell and spatial omics data analysis
- General Statistical Methodologies: High-dimensional model inference and variable selection
- Statistics in Education: Statistical methods for promoting education equity (reported by Forbes)

POSITIONS

Assistant Professor, Michigan State University 08/2025 -
Department of Computational Mathematics, Science & Engineering

EDUCATION

University of California, Los Angeles 09/2020 - 06/2025

Ph.D. in Statistics

Thesis: *Advancing Statistical Rigor in Single-Cell and Spatial Omics Analysis Through In Silico Control Data*

Advisor: Jingyi Jessica Li

Zhejiang University 09/2017 - 03/2020

M.Sc in Probability and Mathematical Statistics

Advisor: Yi Zhang

Shandong University 09/2013 - 06/2017

B.Sc in Mathematics and Applied Mathematics

B.Ec in Economics

GRANTS & AWARDS

National Science Foundation (NSF) Travel Award, STATGEN 2025	2025
Outstanding PhD Student Award, University of California, Los Angeles	2025
Dissertation Year Fellowship (\$38,000), University of California, Los Angeles	2024
JXTX+CSHL Genome Informatics 2023 Scholarship (\$2,000), Cold Spring Harbor Laboratory	2023
Don Ylvisaker Award for the Best Practice of Statistics, University of California, Los Angeles	2023
Interdisciplinary Opportunity Award (\$10,000), NSF-Simons CMCF	2022
Most Promising Statistician Award, University of California, Los Angeles	2022
Summer Mentored Research Fellowship (\$6,000), University of California, Los Angeles	2021
China National Scholarship	2018
Merit Graduate Student Award, Zhejiang University	2018
Outstanding Student Award, Shandong University	2014 - 2016
Hua Loo-Keng Talent Scholarship, Chinese Academy of Sciences	2015

PUBLICATIONS & MANUSCRIPTS

†Indicating co-first author

G. Yan, J.J. Li and M. Biggin (2026). Question-Score Identity Detection (Q-SID): A statistical algorithm to detect collusion groups with error quantification from exam question scores. *arXiv*, 2407.07420. (Accepted for publication in *Patterns*) [Website] [Forbes article] [Podcast]

B. Li, R. Lin, T. Ni, **G. Yan**, M. Burns, J. J. Li and Z. Yao (2025). CellScope: high-performance cell atlas workflow with tree-structured representation. *Nature Communications*, 17, 1130.

E. Huang, T. Fu, L. Zhang, **G. Yan**, R. Yamamoto, S. Terrazas, T.L. Nguyen, C. Gonzalez-Figueroa, A. Khanbabaei, J.H. Bahn, R. Varada, K. Amoah, J. Hervoso, M.T. Paulsen, B. Magnuson, M. Ljungman, J.J. Li, and X. Xiao (2025). Genetic variants affecting RNA stability influence complex traits and disease risk. *Nature Genetics*, 1-11.

Z. Li, Z. M. Patel, D. Song, S. N. Yasa, R. Cannoodt, **G. Yan**, J. J. Li and L. Pinello (2025). Systematic benchmarking of computational methods to identify spatially variable genes. *Genome Biology*, 26, 285.

G. Yan, S. Hua and J.J. Li (2025). Categorization of 34 computational methods to detect spatially variable genes from spatially resolved transcriptomics data. *Nature Communications*, 16, 1141.

J. Zhao, F. Lao, **G. Yan** and Y. Zhang (2024). How data heterogeneity affects innovating knowledge and information in gene identification: A statistical learning perspective. *Journal of Innovation & Knowledge*, 9-3.

G. Yan, D. Song and J.J. Li (2023). scReadSim: a single-cell RNA-seq and ATAC-seq read simulator. *Nature Communications*, 14(1), 7428. [Software] [Website]

D. Song, Q. Wang, **G. Yan**, T. Liu and J.J. Li (2023). scDesign3 generates realistic in silico data for multimodal single-cell and spatial omics. *Nature Biotechnology*, 1-6. [Software]

S. Tang, H. Wang, **G. Yan**, L. Zhang (2022). Empirical likelihood based tests for detecting the presence of significant predictors in marginal quantile regression. *Metrika*, 1-31.

S. Chen†, **G. Yan**†, W. Zhang, J. Li, R. Jiang and Z. Lin (2021). RA3 is a reference-guided approach for epigenetic characterization of single cells. *Nature Communications*, 12(1), 1-13. [Software]

J. Zhao, **G. Yan** and Y. Zhang (2021). Robust estimation and shrinkage in ultrahigh dimensional expectile regression with heavy tails and variance heterogeneity. *Statistical Papers*, 1-28.

J. Zhao†, **G. Yan**† and Y. Zhang (2019). Semiparametric expectile regression for high-dimensional heavy-tailed and heterogeneous data. *arXiv*, 1908.06431. (In press at *Applied Mathematics-A Journal of Chinese Universities*)

PATENTS

M. Biggin, J.J. Li, **G. Yan**. Systems and methods for detecting collusion in student testing using graded scores or answers for individual questions (Serial No. 17/450,984; US Patent 11,915,615 B2)

SOFTWARES

Q-SID An online anti-collusion proctoring system [Website]

scReadSim Python package of synthetic read simulator designed for the single-cell multiomics data [Software]

RA3 R package of “RA3 is a reference-guided approach for epigenetic characterization of single cells” [Software]

PRESENTATIONS

Oral Presentations

- Conference on Statistical Learning and Data Science 2026, New York, NY, USA 11/2026
- ICIBM 2026, Buffalo, NY, USA 08/2026
- STATGEN 2026, Atlanta, GA, USA 05/2026
- Department of Statistics, University of California, Santa Cruz 05/2026
- Department of Statistics, Michigan State University 04/2026
- ENAR 2026, Indianapolis, IN, USA 03/2026
- Biomathematics Graduate Program, North Carolina State University 02/2026
- Department of Computational Mathematics Science and Engineering, Michigan State University 11/2025
- STATGEN 2025, Minneapolis, USA 05/2025
- Jonsson Comprehensive Cancer Center Gene Regulation Seminar, Los Angeles 05/2025
- Department of Computational Mathematics Science and Engineering, Michigan State University 03/2025
- NHGRI Genome Tech Dev Working Group, Jackson Laboratory, USA 01/2025
- Department of Mathematical Sciences, New Jersey Institute of Technology 12/2024
- Department of Statistics and Data Sciences, University of California, Los Angeles 11/2024
- Joint Statistical Meetings, Portland, USA 08/2024
- Institute for Computational and Experimental Research in Mathematics, Providence, USA 12/2023
- Jonsson Comprehensive Cancer Center Gene Regulation Seminar, Los Angeles, USA 11/2023
- Institute for Quantitative and Computational Biosciences Research Seminar, Los Angeles, USA 12/2022
- NSF-Simons Center for Multiscale Cell Fate 5th Annual Symposium, Irvine, USA 10/2022
- The 7th International Conference on Statistics and Probability, IMS-China, Dalian, China 07/2019

Poster Presentations

- Cold Spring Harbor Laboratory Genome Informatics Conference, New York, USA 12/2023
- RECOMB/ISCB Conference on Regulatory & Systems Genomics, Los Angeles, USA 11/2023
- Chan Zuckerberg Initiative Single-Cell Biology 2023 Annual Meeting, Carlsbad, USA 11/2023
- ISMB/ECCB, Lyon, France 07/2023
- Los Angeles Bioscience Ecosystem Summit, Los Angeles, USA 05/2023
- Jonsson Comprehensive Cancer Center Retreat Poster Session, Los Angeles, USA 05/2023
- Institute for Quantitative and Computational Biosciences Poster Session, Los Angeles, USA 09/2022

TEACHING & MENTORING

Teaching

- CMSE 801, Introduction to Computational Modeling, MSU Spring 2026
- CMSE 201, Computational Modeling and Data Analysis, MSU Fall 2025

Teaching Assistant

- STATS 205, Hierarchical Linear Models, UCLA Spring 2024
- STATS 203, Large Sample Theory, UCLA Winter 2024
- Statistical Science with Applications to Epidemiology, ElevatePro Summer 2021
- MATH 1001, Advanced Mathematics, Zhejiang University Fall 2019
- MATH 1001, Advanced Mathematics, Zhejiang University Fall 2018

Workshop Instructor

- Presenter, Jonsson Comprehensive Cancer Center Workshop, UCLA Dec 2024
“Categorization of 34 Computational Methods to Detect Spatially Variable Genes from Spatially Resolved Transcriptomics Data”
- Coordinator & Presenter, QCBio Workshop, UCLA May 2022
“Statistical Methods for Enhancing the Rigor in Single-cell RNA-seq Data Analysis”

Guest Lecturer

- STATS 205, Hierarchical Linear Models, UCLA Spring 2024
- BIOINFO 229, Current Topics in Bioinformatics, UCLA Winter 2024

Graduate Student Mentor

- Huairui Wang, Ph.D. student in CMSE, Michigan State University 08/2026 -
- Feiyu Cao, Ph.D. student in Statistics, Michigan State University 01/2026 -

Undergraduate Student Mentor

- Danny Oh, Michigan State University 06/2026 -
- Abdolla Hegazy, Michigan State University 06/2026 -
- Weijian Wang, Zhejiang University 12/2022 - 06/2023
- Zhiyin Liu, Hong Kong University of Science and Technology 12/2022 - 06/2025
- Shuo Hua, Tsinghua University 06/2022 - 12/2022

PROFESSIONAL EXPERIENCE

Graduate Student Reseracher 09/2021 – 06/2025

Department of Statistics & Data Science
University of California, Los Angeles
Advisor: Jingyi Jessica Li

Research Scientist Intern 06/2024 – 09/2024

Data and Statistical Sciences
Product Development Department
Genentech, Inc.
“Enhancing Statistical Practice in Recurrent Event Endpoint Data Analysis”

Research Scientist Intern 06/2023 – 09/2023

Data and Statistical Sciences
Product Development Department
Genentech, Inc.
“Dynamic Monitoring of Ongoing Clinical Trials for Early Decision Making”

Research Assistant 06/2019 – 04/2020

Department of Statistics
The Chinese University of Hong Kong
Advisor: Zhixiang Lin

Undergraduate Researcher 06/2016 – 06/2017

Academy of Mathematics and Systems Science
Chinese Academy of Sciences
Advisor: Jianming Xia

PROFESSIONAL SERVICES

Conference and Session Organization

- Session Organizer, STATGEN 2025 Conference May 2025
“Spatially Variable Gene Detection: Advances and Applications in Spatial Transcriptomics data”

Statistical Consulting Services

- Jonsson Comprehensive Cancer Center 2022 - 2024

REVIEWING ACTIVITIES

Reviewer for Scientific Journals

Nature Communications, BMC Bioinformatics, BMC Genomics, BMC Methods, Nucleic Acids Research, Genome Biology, Biometrics, PLOS Computational Biology, Cell Reports, Patterns, Bioinformatics Advances, Statistics in Medicine, Genome Research, Review Commons, Physiological Genomics, Nature Naunyn-Schmiedeberg's Archives of Pharmacology, npj Artificial Intelligence, Bioinformatics

Reviewer for Scientific Conferences

Eastern North American Region of the International Biometric Society (IBS) Conference, ACM BCB 2025 Conference, 2025 NCME Annual Meeting

PROFESSIONAL AFFILIATIONS

NSF-Simons National Institute for Theory and Mathematics in Biology	2025 - Present
Institute of Mathematical Statistics	2023 - Present
American Statistical Association	2022 - Present
American Society of Human Genetics	2022 - 2023
UCLA Jonsson Comprehensive Cancer Centers	2022 - Present
International Indian Statistical Association	2022 - 2023